

LAB 02 REPORT - SDN302

Requirement 1. Explore a public REST API with Postman (20%)

#	Method	URL	Request Body	Status Code	Response Headers quan trọng
1	GET	https://jsonplaceholder.typicode.com/posts	Không có	200 OK	Content-Type: application/json; charset=utf-8 Content-Length: 27520 Cache-Control: max-age=43200 X-Powered-By: Express Connection: keep-alive
2	POST	https://jsonplaceholder.typicode.com/posts	json { "userId": 1, "title": "foo", "body": "bar" }	201 Created	Content-Type: application/json; charset=utf-8 Location: /posts/101 Content-Length: 63 X-Powered-By: Express Cache-Control: no-cache
3	PUT	https://jsonplaceholder.typicode.com/posts/1	json { "id": 1, "userId": 1, "title": "updated	200 OK	Content-Type: application/json;

			title", "body": "updated body" }		charset=utf-8 Content-Length: 84 X-Powered-By: Express Cache-Control: no-cache
4	DELETE	https://jsonplaceholder.typicode.com/posts/1	Không có	200 OK	Content-Type: application/json; charset=utf-8 Content-Length: 2 X-Powered-By: Express Cache-Control: no-cache

JSON to XML example

Example JSON response:

```
{"userId": 1, "id": 1, "title": "sunt aut facere repellat provident occaecati excepturi optio reprehenderit", "body": "quia et suscipit..."}
```

Equivalent XML structure:

```
<post><userId>1</userId><id>1</id><title>sunt aut facere repellat provident occaecati excepturi optio reprehenderit</title><body>quia et suscipit...</body></post>
```

Two advantages of JSON over XML

1. JSON is shorter and easier to read than XML.
2. JSON is more convenient for JavaScript and web applications because it is easy to parse and use.

SOAP vs REST

SOAP:

- Protocol: SOAP is a protocol with strict standards.
- Message format: Usually XML.

- State: Often stateful.
- Typical use case: Enterprise systems, banking, and service integration.

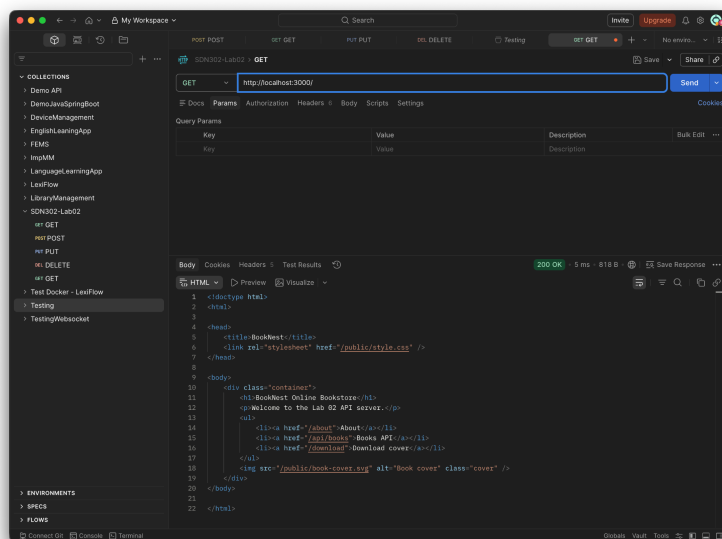
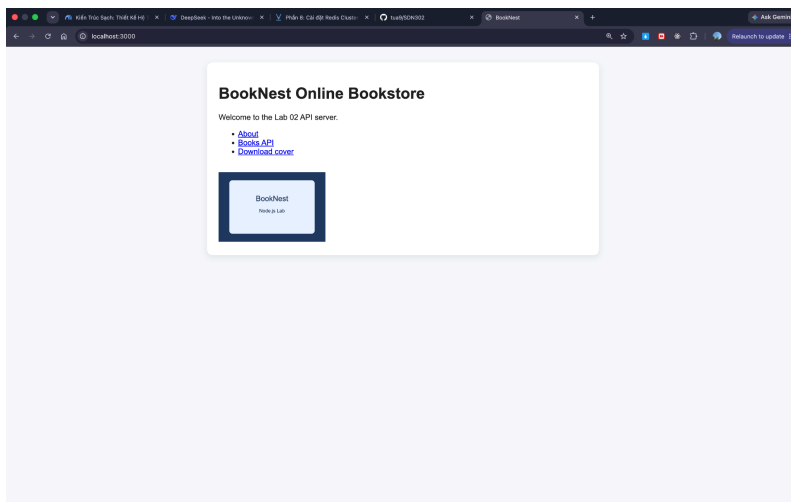
REST:

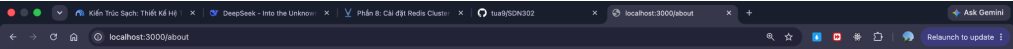
- Protocol: REST works over HTTP and uses standard methods like GET, POST, PUT, DELETE.
- Message format: Usually JSON, but can also use XML.
- State: Stateless by design.
- Typical use case: Web APIs, mobile apps, and modern microservices.

Exported Postman collection: SDN302-Lab02.postman_collection.json

Requirement 2. Build a web server with the core http module (25%)

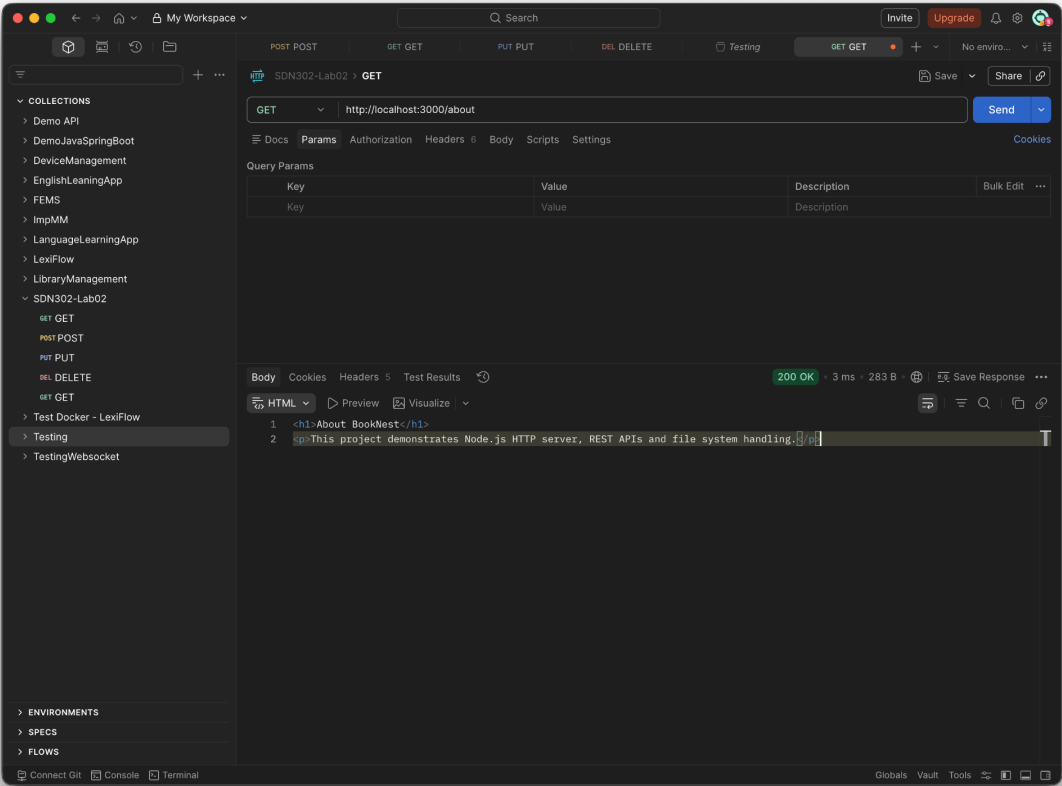
Path: /, /about, /api/books in browser and postman





About BookNest

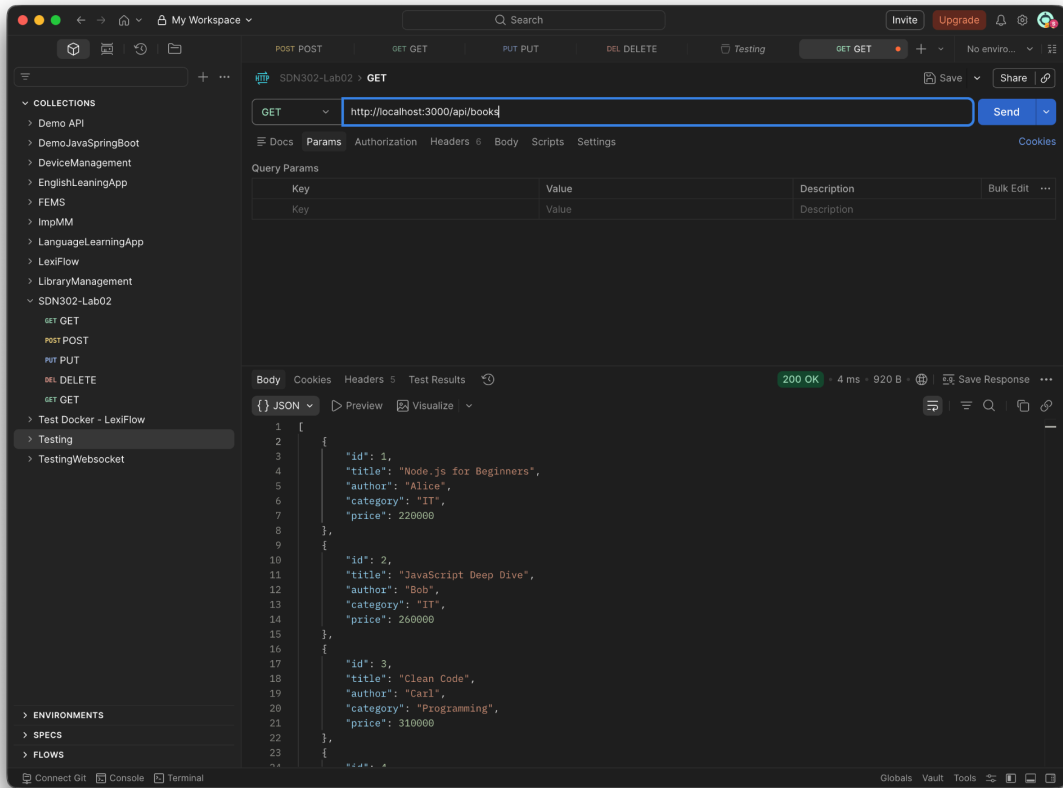
This project demonstrates Node.js HTTP server, REST APIs and file system handling.



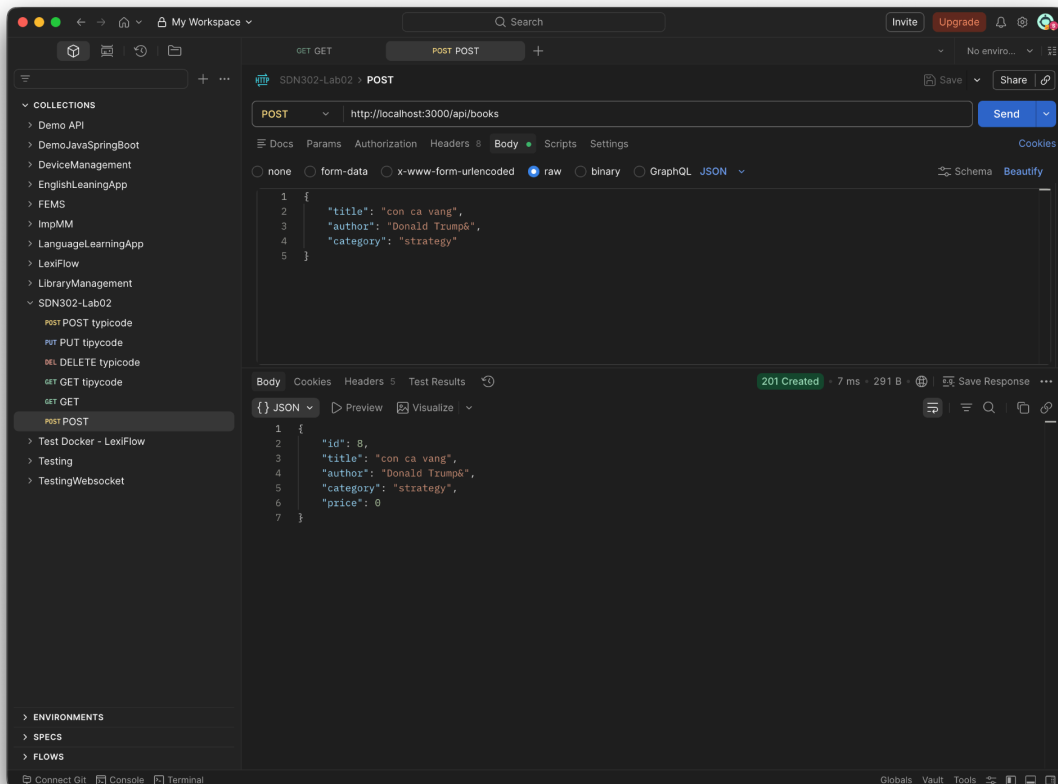
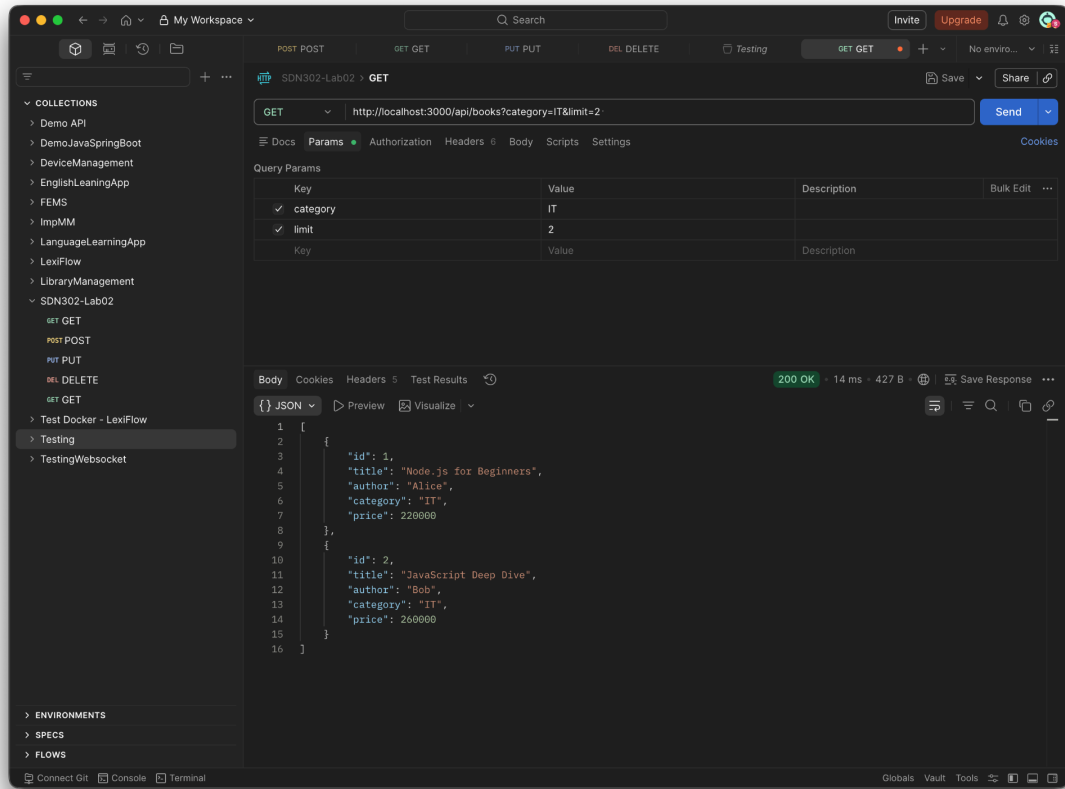
```
localhost:3000/api/books

Pretty-print

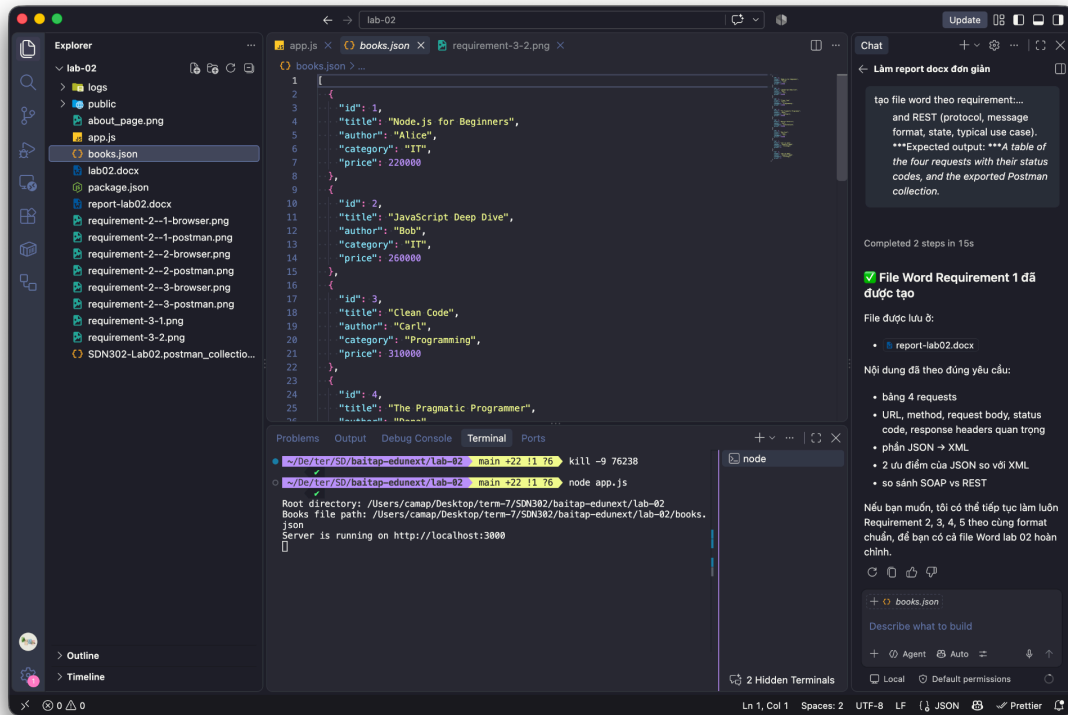
{
  "id": 1,
  "title": "Node.js for Beginners",
  "author": "Alice",
  "category": "IT",
  "price": 220000
},
{
  "id": 2,
  "title": "JavaScript Deep Dive",
  "author": "Bob",
  "category": "IT",
  "price": 260000
},
{
  "id": 3,
  "title": "Clean Code",
  "author": "Carl",
  "category": "Programming",
  "price": 310000
},
{
  "id": 4,
  "title": "The Pragmatic Programmer",
  "author": "Dana",
  "category": "Software",
  "price": 290000
},
{
  "id": 5,
  "title": "Design Patterns",
  "author": "Eve",
  "category": "Architecture",
  "price": 350000
},
{
  "id": 6,
  "title": "New Book",
  "author": "Test",
  "category": "IT",
  "price": 150000
}
}
```



Requirement 3. Handle HTTP methods, query strings and request bodies (20%)



Requirement 4. Work with the File System (25%)



Requirement 5. Serve static files (10%)

